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50TH ANNIVERSARY ARTICLE

The Attentional Blink: Then and Now

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The attentional blink (AB) refers to the outcome where in a rapidly presented stream of stimuli observers are unable to successfully report the second of two previously defined targets relative to an otherwise identical control condition where the first target does not require report. The first AB article was published in the *Journal of Experimental Psychology: Human Perception and Performance* in 1992 and in the ensuing 32 years has garnered the interest of many researchers spanning multiple disciplines and experimental approaches, both basic and applied. Moreover, it has attracted the attention of a variety of social media whose authors have appropriately and in some cases inappropriately used the AB to account for human behavior. Perhaps one of the most important outcomes of this research has been to shift the focus from the study of spatial attention to include that of attention over time, or temporal attention. Given the upcoming 50th anniversary of *Journal of Experimental Psychology: Human Perception and Performance*, the journal has asked the original authors to reflect on the history of attentional blink research. We wish to acknowledge that such reflection must be bounded by space and time (pun intended) limitations so apologies to those authors whose undoubtedly important research we were unable to cover.

Public Significance Statement
The original attentional blink article published in *Journal of Experimental Psychology: Human Perception and Performance* in 1992 reveals that the report of two consecutive targets is impaired when separated by a short temporal interval. The present report is an overview of the literature stemming from this article, reviewing some of the key findings and demonstrating its impact and reach across multiple scientific disciplines.

Keywords: attentional blink, visual information processing, auditory information processing

In 1992, the *Journal of Experimental Psychology: Human Perception and Performance* published the first peer-reviewed attentional blink (AB) article (Raymond et al., 1992). The purpose of the present article is to briefly characterize the history of the AB research that was sparked by the publication of this article and to highlight the main contributions and impact of AB research more generally. In addition to the seminal AB article, many of the key papers that shaped AB research were published in the *Journal of Experimental Psychology: Human Perception and Performance*. With this article, we celebrate the Journal's 50th anniversary with a brief, informal review of the last 30 years of AB research.

The seminal AB article, referred to here as the "1992 article," demonstrated and described how the capacity to identify a simple, briefly presented visual target could be markedly impaired by imposing another visual task just prior to the target's appearance. Engaging with one stimulus seemed to temporarily shut down awareness of subsequent stimuli. Coined the "attentional blink" by Jane Raymond in a 1991 Psychonomics Society poster, this phenomenon demonstrated that people could experience a transient loss in awareness while awake and focused on a task. The 1992 article reported four experiments that initiated the interrogation of this interesting, robust effect. Considering the substantial theoretical

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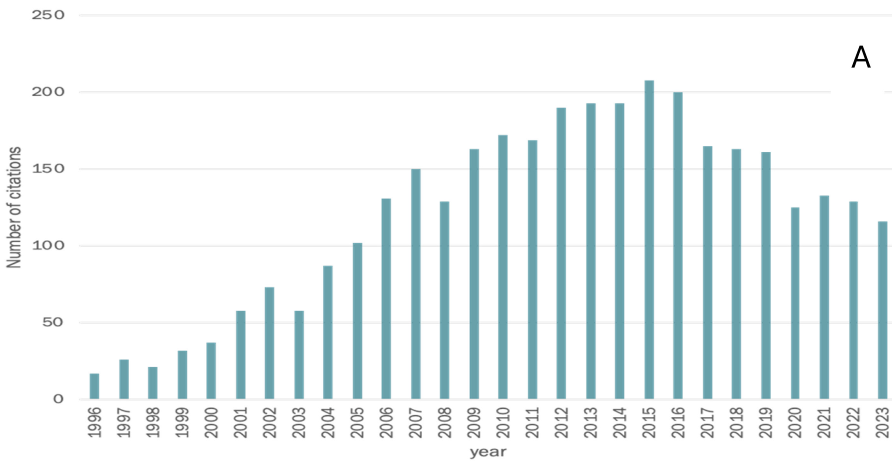
Figure 1

Citations of Raymond et al. (1992) From 1996 to 2023 as Reported by Google Scholar



Note. See the online article for the color version of this figure.

and practical implications of the effect, interest was quickly sparked, leading to intense investigation during the subsequent 30 years by cognitive, neuro-, and vision scientists from around the globe.

Indeed, the 1992 article has been cited well over three thousand times (Google scholar) and has led to nearly 100 PhD theses. Figure 1 shows citation rates over the last 30 years, showing that the 1992 article reached its peak influence by this metric around 2012, 20 years after its publication. There are arguably three main reasons why the AB had such a substantial impact. First, it addressed issues of broad interest to cognitive psychologists and neuroscientists that concern the complex relationships among perception, short-term (or working) memory, long-term memory, and consciousness. Despite the titular reference to vision, the AB has been observed in other sensory modalities, underscoring its general importance to cognitive and sensory psychology. Second, it used a simple, elegant methodology that could be flexibly applied to many different research questions and approaches. Third, the AB had obvious practical implications for cognition in everyday life, making it readily accessible as a cognitive psychology concept for students and the public.

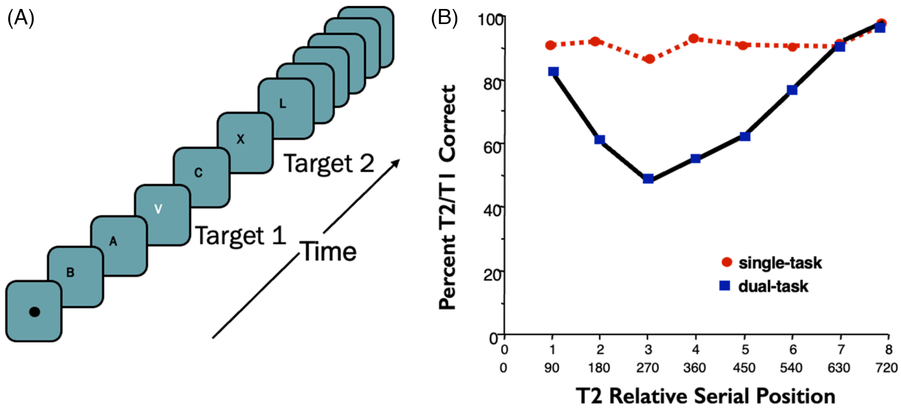
We begin with a brief summary of the main findings from the 1992 article and an outline of the theories and issues that arose within the first decade or so. Next, we outline the main approaches to and motivations for research that ensued over the succeeding two decades. We finish with a summary of the impact the AB has made on related fields within psychology and neuroscience, on academic areas outside of Psychology, and on applied arenas including the law, the marketplace, and popular social-media culture.

The 1992 Article and the Early Years of AB Research

The studies leading to the 1992 article had their antecedents in the study of spatial attention. The predominant view in the late 1980s and early 1990s was that attention operated like a spotlight, facilitating perception of a spatially localized area of the visual field (Posner, 1980). It was well established by that time that moving attention to different areas of the visual field first required a disengagement process, movement, and then reengagement at the new location. A primary implication of this view of attention is that high level processing of different objects in a static scene is serial and takes time (Treisman & Gelade, 1980). However, there was little understanding of how attention operates when objects in the visual field abruptly change appearance but not location as, for example, in a video scene. We reasoned that if attention functions to boost processing of relevant features and objects, there could be a disengagement cost when attention has to be terminated for one object and then immediately engaged for another, even when no spatial shift in attention was required. Indeed, supporting this possibility were several articles published in the 1970s and 1980s revealing a processing "refractory period" when multiple, serially presented items were displayed at the same visual field location, a technique known as rapid serial visual presentation (RSVP; see Broadbent & Broadbent, 1987 for a review). These studies presented relatively complex stimuli, demanded relatively complex responses, and offered diverse explanations for their results. Some looked to sensory processes, for example, masking, to explain recognition and recall errors, whereas others referred to memory and categorical interference processes. Raymond et al. (1992) provides a more comprehensive description of these antecedent reports for the interested reader. Our aim in the 1992 article was to simplify the procedure for assessing temporal effects on the processing of successive visual stimuli and to explore more directly the possibility of an attentional basis for these previously observed constraints.

The methodology we developed is illustrated in Figure 2 and was subsequently widely adopted, forming the basis of the procedures used in numerous highly influential articles (e.g., Chun & Potter, 1995). On each of a series of trials, a stream of letters is presented at fixation (Figure 2A). Each item is presented for 15 ms with an interstimulus interval (ISI) of 75 ms, yielding a 90-ms stimulus onset asynchrony (SOA). On AB trials, the series includes two "targets" inserted into a series of nontargets (distractors). The first target (T1) is a uniquely white letter in a stream of black letters and can be any letter of an alphabet subset except the letter "X." The second target (T2) is a black letter "X" and is only presented on a randomly selected 50% of trials (AB trials). Participants are requested to report the identity of T1 and report whether T2 is present or absent at the end of each letter series. Between eight and 15 distractors are

Figure 2
Stimuli and Results From Raymond et al. (1992)



Note. (A) An illustration of the stimuli used in Experiment 2 of Raymond et al. (1992). (B) The results of Experiment 2. The top numbers of the *x* axis reflect the lag position; the bottom numbers reflect T1/T2 stimulus onset asynchrony. T1 = first target; T2 = second target. See the online article for the color version of this figure.

presented prior to T1 but eight letters always follow T1. On AB trials, T2 can appear equally often at any one of the post-T1 serial positions. In a control condition, participants ignore the T1 task and just report the presence versus absence of T2. Importantly, in this condition, the perceptual input remains the same as in the dual-task condition, but the cognitive demand is different.

The main measure of interest is performance accuracy for T2 on AB trials in the dual-task condition, conditional on correct responses for T1. The classic result which now has been replicated many times is that T2 accuracy drops substantially when T2 is presented 300–600 ms after T1. This drop is the AB. Its magnitude has been calculated in numerous ways but most commonly as the maximum drop relative to later T2 presentations (at least after 800 ms after T1). However, in the 1992 article, T2 performance on the single- and dual-task conditions was directly compared, as shown in Figure 2B, as way of characterizing the AB. A significant interaction of trial condition (dual vs. single task) by T1–T2 interval is used to evidence the AB. It is important to note that the procedure used in the Raymond et al. report was modeled after a prior experiment by Weichselgartner and Sperling (1987). In their study, participants were required to monitor a stream of RSVP digits presented at a single location, with the target (T1) digit denoted by a box around it. They were then required to report as many digits including and following the target as possible. The results found that participants were able to report the target, the T1 + 1 item but then missed a few items before recommencing accurate report. Although this report found an outcome like that of the AB, there was no control group to enable the authors to conclude that the outcome was because of early perceptual processes or later, for example, attentional, accounts. Moreover, the procedure involved a substantial memory load, thus complicating interpretation.

We think it is worth sharing anecdotally the enthusiasm pervading our lab (in which Karen Arnell was a research assistant) in the early days of research on this topic. We recall that the excitement at lab meetings was "palpable," leading to a long list of possible experiments on the table. Even though there were clear-cut avenues we wished to pursue, we quickly realized that we could not follow them all. So, we decided to make the computer code for running an AB experiment readily available to anyone who requested it, helping others to explore our exciting finding, even though this was many years before "open science" was a recognizable term! Although there never are control groups in real life, we remain convinced to this day that this was a good decision and contributed substantially to the steep rise of research into the AB.

The phenomenon of the AB sparked many studies, and with them, several theories were developed about its cause. Most centered on the general notion that the AB stems from a resource limitation imposed by the demands of processing T1, an idea that was consistent with prevalent theories of spatial attention. Our initial explanation of the AB in the 1992 article involved a two-stage inhibitory model. We proposed that T1 detection required an initial, rapid detection stage that was followed by a slower and more demanding process that enabled conscious T1 identification. The second stage was thought to provoke suppression of any subsequent processing so as to prevent T1 errors (e.g., misidentifying a subsequent item as the T1). This suppression was thought to extend in time, causing a temporary decrement in T2 accuracy, aka the AB. However, subsequent experiments revealed a robust AB even when the immediate post-T1 item was highly dissimilar to T1 (Chun & Potter, 1995; Raymond et al., 1995) and when T1 required only a simple detection task, for example, "was there a white item or not" (Shapiro et al., 1994). Moreover, the suppression notion suggested that T2 was not processed at all, whereas successive experiments showed that T2 could prime a subsequent item, indicating that it had been internally represented but was not consciously available (Shapiro, Driver, et al., 1997). To explain the priming and other related results regarding the critical role of the items appearing immediately after both T1 and T2, we then put forth an interference model based on the notion of competition in visual short-term memory as proposed by Duncan and Humphreys (1989). Interference theory proposed that T1, T2, and their respective masks are processed to a varying degree and that they compete for retrieval from a working memory storage buffer. An extension of the same idea was the basis of the dwell-time model proposed later by Duncan and colleagues (Duncan et al., 1994; Ward et al., 1996). The reader is referred to Dux and

Marois (2009) for a detailed review of these and related models of the AB and to Denison et al. (2021) for a recent review of temporal attention more generally.

Behavioral Studies of the AB

Behavioral research began to proliferate about 10 years after the first AB article was published. These studies varied the nature of the stimuli (presenting letters, words, faces, objects, or scenes), their physical features, their duration, the presence of other distracting stimuli, the nature of the first and second tasks (sometimes adding additional tasks), the spatial location of both targets and distractors, and even the number of concurrently presented streams of items. Some of this research was directed at investigating the features of the AB phenomena itself exploring, for example, modality specificity, lag-1 sparing effects, or individual differences. Other work was directed at understanding links between the AB and other sensory, cognitive, and psychological phenomena, with the aim of understanding the mechanisms causing the AB. Here, we provide a brief but by no means exhaustive outline of this work.

Modality Specificity

An obvious question relating to the possibility of an attentional basis for the AB was whether it is specific to visual processing, could operate within other sensory modalities, or operates at a higher, amodal level. An AB in the auditory domain has been shown with many of the same manipulations, for example, masking or variations in the T1–T2 interval, revealing effects like those in the visual domain (Arnell & Larson, 2002; Mondor, 1998; Tremblay et al., 2005; Vachon & Tremblay, 2005). Whereas most studies used a behavioral approach, a few have used an electrophysiological marker (P300) to index the presence of an auditory AB (Arnell, 2006). A few studies examined the AB in the tactile domain where an AB was found only when a location judgment was required but not when intensity, duration, or frequency judgments were required (Hillstrom et al., 2002). However, evidence of a cross-model AB (T1 presented in one modality and T2 in another modality) has remained equivocal. At least one study failed to find a visual-auditory AB (Koelewijn et al., 2008), whereas others have reported this effect (Arnell & Jolicœur, 1999; Potter et al., 1998). Moreover, a visual-tactile AB effect has also been reported, supporting the notion of a cross-modal or amodal process (Soto-Faraco et al., 2002). One study examining the visual AB reported no AB when the T1 task was to judge the temporal duration of a gap in the visual stream as "short" or "long" and a letter T2 had to be identified, as in a typical AB experiment (Sheppard et al., 2002). Having to identify the same duration (T1) letter reinstated the AB. This result also supports the possibility of an amodal AB process.

Lag-1 Sparing

"Lag 1 sparing" refers to the experimental finding that the AB typically does not start, that is, reveal a difference between the experimental and control conditions (see above) until ~150 ms after T1. This means that the T1 + 1 (i.e., the first item following T1) RSVP stream item is "spared" from the AB, yielding a U-shaped curve when T2 performance is plotted as a function of T1–T2 interval. This outcome has been witnessed in many AB experiments, including the original 1992 report. Various accounts were advanced to explain lag-1 sparing (e.g., Visser, 2015), and various theories of the AB were supported or rejected based on conditions that either gave rise or not to this effect (e.g., Kimata et al., 2023). Importantly, if a stimulus item is inserted into the RSVP stream immediately following T1 but before the T1 + 1 (mask) item occurs, lag-1 sparing of the T1 + 1 item is no longer evident (Drew & Shapiro, 2006). This suggests that the T1 + 1 item (T1 mask) is a factor causing the AB but is not manifest in the expression of the AB, per se.

Individual Differences

As is frequently the case with experimental reports, and the AB is no exception, outcomes and therefore accounts to explain them are based on data averaged across participants. Whereas averaged data typically yield more reliable and valid conclusions, it notoriously hides interesting individual differences that in some cases intelligently inform theories. Indeed, one of the first articles to report individual differences in the AB revealed large discrepancies in performance among participants (Klein et al., 2011). Following on from this, Martens et al. (2006) used event-related potential methodology to reveal that "nonblinkers" showed an earlier P3 peak, suggesting that they were able to consolidate targets faster than "blinkers." From this stemmed many subsequent articles examining AB effects in special populations, a point addressed in more detail below.

Related Effects

Paralleling studies of the size and nature of the AB effect, per se, were many studies attempting to link the AB with other known phenomena and to identify its underlying mechanisms. Perhaps the most intensively studied was the link between the AB and masking. Masking is a well-studied perceptual outcome revealing effects on detection and/or identification of a target stimulus resulting from a second stimulus appearing closely in time before (forward masking) or after (backward masking) the target. Examining the effects of low-level masking, Seiffert and Di Lollo (1997) confirmed a previous finding that the AB is reduced when the T1 mask is omitted, and then went on to show that the AB is increased when superimposing the T1 mask with T1 but reduced when the T1 mask does not spatially overlap. Several groups, including our own, studied the AB with the aim of exploring how sensory masking effects might contribute to the AB (e.g., Brehaut et al., 1999; McLaughlin et al., 2001; Raymond et al., 1995). Such studies manipulated the items immediately following T1 and T2 in terms of their similarity to target items in visual (e.g., color, size, and form), categorical, temporal, and spatial characteristics. The general aim was to match the temporal and contrast functions known to characterize sensory masking effects with AB effects. The general outcome was that sensory or perceptual masking cannot fully explain the AB, although masking effects clearly impact conscious perception (Kouider & Dehaene, 2007). A related line of research explored the level of item representation that led to interference in a putative working memory buffer. Two studies using the AB provide evidence that interference concerns object-level representations, rather than feature-specific information (Raymond, 2003; Sheppard et al., 2002) by demonstrating that the AB occurs when tasks require the identification of new objects, not new features. This work also argues against a masking basis for the AB and even efforts to explore whether object-level masking (aka object substitution masking) underpinned the AB

(Giesbrecht & Di Lollo, 1998) eventually rejected this possibility (Giesbrecht et al., 2003).

Spatial and Temporal Attention

An integrated understanding of spatial and temporal attention has been a key goal for many AB studies. An important contribution on this point was made by Duncan and colleagues when they presented two targets in different but predefined spatial locations and found a characteristic AB (Duncan et al., 1997; Ward et al., 1996). The conclusion was drawn that, although conventional models of attention when stimuli are presented concurrently reveal identification following a serial function and taking ~30–40 ms per item, serially presented stimuli reveal a much longer "dwell time" for correct identification. In another study examining spatially separated stimuli in an RSVP stream, Kristjánsson and Nakayama (2002) presented six streams of letters with targets occurring in any of the streams to find that more distally presented streams yield less AB than those presented more proximally. Although studies introducing spatial separation between serially presented stimuli are useful in elucidating the interaction between spatial and temporal attention, they must be interpreted with caution when compared to the typical AB method of nonspatial presentation as they introduce a new element (space) that may have both additive and interactive effects.

The AB and Other Temporal Attention Effects

A large body of research aimed to draw connections with other experimental paradigms showing effects of temporal attention. Connections have been drawn between the AB and the psychological refractory period (PRP; Dell'Acqua et al., 2001; Marti et al., 2012), repetition blindness (RB; Arnell & Shapiro, 2011; Chun, 1997; Dux & Marois, 2007; Jiang et al., 2013), biased competition (Ward et al., 1996), inattentional blindness (Beanland & Pammer, 2012), contingent attentional capture (Folk et al., 2002; Zivony & Lamy, 2016), object substitution masking (Giesbrecht et al., 2003), and emotion-induced blindness (Singh & Sunny, 2023). There is little doubt that there are similarities among the above phenomena and the AB and likely a significant overlap in the cognitive processes underlying each. However, as the expression goes, the devil is in the detail; in this case, of the subtle but important conclusions that can be drawn from the application of one approach relative to the other. Two examples, one concerning RB and the other the PRP, illustrate this point.

The AB and RB employ virtually the same method of requesting a participant identify two targets, separated by a varying SOA of between approximately 100 and 700 ms. The outcome of both is that the second target often fails to be reported correctly or, dramatically in the case of RB, fails even to be detected despite in the case of reading a sentence, relegates it to a grammatical failure. What differentiates AB from RB is that, in the latter, the two targets are identical or highly, for example, semantically, related; in the former, the targets may be from the same category, for example, letters, but are not necessarily related and in many cases drawn from different categories, for example, digits and letters. The two effects have different time courses and can be double dissociated by manipulating target-distractor similarity and episodic distinctiveness (Chun, 1997). Importantly, the accounts purporting to explain RB and AB, though many and varied, converge generally to argue that the second target in both AB and RB experiments is processed to an early stage in the processing hierarchy but fails to be encoded to a (later processing) level suitable to enable the report required by the experiment in question.

Although superficially similar at the level of stimulus presentation, the AB and the PRP have an important difference in report requirement. The former requires report of both targets at the termination of the presentation of the stimulus stream, whereas the latter requires report immediately after each target is presented. Although some investigators have argued the outcome of both AB and PRP is because of a central processing bottleneck (Dell'Acqua et al., 2001), there is a strong argument to be made that this terminology is too broad and fails to reflect important differences in cognitive processing and the underlying neural correlates to be useful.

The AB and Nonsensory Stimulus Characteristics

Numerous studies have sought to understand how nonsensory stimulus characteristics, for example, learned associations, familiarity, or emotional valence, might influence stimulus representational strength in a way that could be revealed by the AB. Indeed, several studies have shown that target items with emotional associations such as emotional words (Huang et al., 2008) or taboo words (Mathewson et al., 2008) mitigate AB effects. Other studies reveal how T2 items with high familiarity "escape" the AB, even when complexity is controlled (Jackson & Raymond, 2006). In a study that directly manipulated individual reward history, Raymond and O'Brien (2009) showed reduced AB effects for T2 item that had been previously paired with reward but were otherwise emotionally neutral. Collectively, these studies of emotional and motivational salience support the interference theory account of the AB by suggesting that stronger, highly relevant targets reduce the AB effect, especially when they are applied to the second target.

The AB and Consciousness

The AB is widely considered as a lapse of consciousness and numerous studies have used it as an approach for exploring the nature of consciousness, the representation of information in the brain, and the neural correlates of consciousness (Tononi & Koch, 2008). For example, Dehaene et al. (2006) employed the AB as an experimental tool to examine under what conditions a stimulus becomes conscious, and then used their results to develop their theory of a global neuronal workspace. Similarly, Shapiro and his colleagues used a deep convolutional neural network approach in a categorial AB paradigm to examine how mid- to high-tier areas of the brain provide object-specific access to conscious awareness of natural image (Lindh et al., 2019).

The AB and Cognitive Control

Another focus of AB research is related to cognitive control over the task. Several reports advanced a theory of the AB known as temporary loss of control (TLC; Di Lollo et al., 2005; Kawahara, Enns, & Di Lollo, 2006; Kawahara, Kumada, & Di Lollo, 2006) where it is argued that the AB is caused by disruption to a control mechanism that deals with the target tasks. According to TLC, task switches are demanded when the T1 and T2 tasks require different selection modes, which is posited as the underpinning AB mechanism. In TLC experiments, typically three targets are presented with a variable interval between them. Targets can be of the same category, for example, all letters, or different categories, for example, letters and digits. The results of such experiments generally reveal that the former condition yields either no or a reduced AB, whereas the latter reveals the typical AB outcome. Of particular interest is the finding that under certain circumstances, even three targets fail to reveal an AB. Although this account is compelling, there is at least one condition in Shapiro et al. (1994) that is discrepant with the above account. In Experiment 3B, participants were requested to search for the presence/absence of the black letter "S" as T1 then do the identical presence/absence task for the black letter "X" as T2. Although this experimental condition could be argued to produce a TLC given that both targets are black letters, a substantial AB was observed. Nevertheless, issues of shifts in task selection remain an interesting avenue of AB research.

Other Research Approaches

Research approaches to study the AB have spanned the gamut. These include neuroimaging, electrophysiology, computational modeling, and clinical studies of special populations. Here we briefly highlight AB research efforts made with each approach and the key issues they addressed.

Neuroimaging

Functional brain neuroimaging approaches to the study of the AB have been relatively few but those that have been conducted generally support the view that T2 information is represented in the brain during the AB but fails to reached a level of processing sufficient to enable conscious report. Two studies (Marois et al., 2004; Shapiro et al., 2007) used functional magnetic resonance imaging to reveal that the second target (T2), even when it failed to be reported, nevertheless produced brain activation in posterior parietal cortex, though to a significantly lesser extent than when T2 is reported correctly. Moreover, increased activity in frontal and parietal regions, such as prefrontal cortex and intraparietal sulcus, is associated only with successful target detection.

Electrophysiology

Studies using an electrophysiological approach are more numerous than those employing neuroimaging and have produced largely similar findings. One of the first electrophysiological studies of the AB (Luck et al., 1996) showed that early sensory cortical indices (P1, N1), as well as a later semantic component (N400), are equally present on both AB (T2 not detected) and non-AB trials (T2 detected), showing remarkably that semantic processing survives the AB. In contrast, a component occurring approximately 300 ms posttarget (P300), which indexes when a stimulus has been stored in working memory (aka conscious awareness), is present only on trials when an AB does not occur. Findings such as these strongly suggest that the AB is not a failure to process a stimulus but a failure to transfer it to a durable form of working memory.

Electrophysiological studies have also led to interpretation of the AB in terms of a neural oscillatory framework. Nearly all behavioral studies of the AB have used timing parameters of 15 ms on/75 ms off for each stimulus item in the letter stream, yielding a presentation rate of 90 ms per item or 11.11 items per second; a rate within the alpha frequency band (8–12 Hz). Alpha is one of the most robust and frequently studied neural oscillations and typically reflects behaviorally relevant neural states (Linkenkaer-Hansen et al., 2001). Studies have revealed that accurate perception of a visual stimulus can be predicted by the amplitude of prestimulus alpha (van Dijk et al., 2008). The role in the AB of a functionally related frequency band to alpha, that is, beta, was examined by Gross et al. (2004) and found to have network properties that significantly influenced AB magnitude. Following on from this, Shapiro et al. (2017) found that altering the RSVP presentation rate to a frequency either slower (theta) or faster (gamma) than alpha significantly attenuated AB magnitude, in turn revealing the importance of this parameter in the study of the AB.

Computational Models

There are several computational models of the AB that have garnered considerable attention over the years and here we focus on three. The first modeled the time course of the locus coeruleus (LC) to advance the position that the response of this brainstem nucleus may be involved in the AB (Nieuwenhuis et al., 2005). In this report, the authors note the LC has been shown to phasically release norepinephrine when processing motivationally relevant stimuli, which in turn increases the gain on efferent target neurons to facilitate stimulus processing. Another model developed to explain the AB was proposed by Wyble, Bowman, and colleagues and is referred to as the "simultaneous type, serial token" (ST^2) model (Bowman et al., 2022; Bowman & Wyble, 2007; Wyble et al., 2009, 2011). In its most simplistic form and consistent with the basic tenant of the LC model, ST^2 argues that humans temporally sample the world with interruptions of nonsampling as revealed by the AB. The model is manifest in five fundamental principles: (a) It is based on the two-stage model advanced by Chun and Potter (1995) that describes a parallel processing stage followed by a serial stage; (b) in the parallel stage competing stimulus items are filtered according to salience; (c) types (featural representations) versus tokens (episodic context) are distinguished; (d) a mechanism provides a brief enhancement of the type representation; and (e) a binding pool associates types with tokens. In support of their model, it was found that sampling four targets presented in an uninterrupted sequence yields more accurate report than the same four targets presented with intervening nontargets, though at a cost of decreased temporal order precision (Olivers et al., 2007). A third model is the Boost and Bounce model of Olivers and Meeter (2008). Here the notion of capacity limitations or bottlenecks are eschewed and, instead, the operation of a hypothetical gating system that can enhance or suppress relevant or irrelevant information, respectively, is proposed. The model in essence suggests that both the (first) target and the distractor following this target are enhanced (boosted) but the distractor is subsequently inhibited (bounced), in turn causing the subsequent inability to identify items for the period of time referred to as the AB.

Clinical Approaches

Many studies of clinical disorders that are associated with cognitive or perceptual difficulties have used the AB to assess potential problems of attention. As far as the present authors are aware, there has not been a review article on clinical approaches and the AB. However, Table 1 provides a nonexhaustive summary of the more widely studied clinical groups/disorders. Most of these reports conclude that a larger AB is evident in the clinical group under investigation relative to the appropriate control, though in some cases, for example, smoking, this result is reversed, that is, there is a decreased AB in the clinical group.

Table 1
Examples of Clinical Studies Using the Attentional Blink

Attention deficit hyperactivity disorder (Amador-Campos et al., 2015; de Groot et al., 2015; Donnadieu et al., 2015; Zhang et al., 2023)	Autism (English et al., 2017; Gaigg & Bowler, 2009; Nijhof et al., 2020, 2022; Xu et al., 2015)
Alzheimer's disease (Perry & Hodges, 2003; Vila-Castelar et al., 2019)	Brain lesions, stroke (Anderson & Phelps, 2001; Correani & Humphreys, 2011; Li et al., 2016)
Cannabis (Campbell et al., 2018)	Dyslexia (Badcock et al., 2008; Buchholz & Davies, 2007; Facoetti et al., 2008)
Eating disorder (Arumäe et al., 2019; DePalma et al., 2017; Neimeijer et al., 2017; Schmitz et al., 2015; Woltering et al., 2021)	Gambling (Farr et al., 2024)
Obsessive–compulsive disorder (Olatunji, 2021)	Parkinson's disease (Slagter et al., 2016)
Pain (Kreddig et al., 2022)	Trauma and posttraumatic stress disorder (Krans et al., 2012; McHugo et al., 2013; Olatunji et al., 2022)
Schizophrenia (Jimenez et al., 2016; Strauss et al., 2013; Su et al., 2015; Walsh-Messinger et al., 2014)	Smoking (Waters et al., 2007)
Spider phobia (Trippe et al., 2007)	Stress (Kan et al., 2019; Olatunji et al., 2022; Schwabe & Wolf, 2010)
Subclinical depression (J. Wang, Luo, et al., 2022)	Substance use disorder (Lu et al., 2024)

Applied, Nonclinical Impact, and Social Media

The AB method has been adapted to investigate a wide range of nonclinical, applied problems (see Table 2 for examples). Some of this research has been motivated by a need to understand constraints on awareness in everyday situations and, specifically, how and when obvious information is missed in dynamic, ongoing situations, such as driving (e.g., Hashemirad et al., 2023), watching video advertisements (e.g., Shen et al., 2020), or listening to music (e.g., Jefferies et al., 2008; Olivers & Nieuwenhuis, 2005). Other work has focused on using the AB to index the strength of internal representations of specific stimuli. Here, the general notion is that a reduced AB indicates that stimuli presented as T2 are more strongly, or more readily represented. For example, studies of sex offenders have shown a reduced AB for sex-related stimuli compared to controls (e.g., Beech et al., 2008). Similarly, certain types of lying (e.g., claiming not to know a person) have been revealed by smaller than expected AB effects when that person's image is presented as T2 (Bowman et al., 2014). Using similar logic, studies of bilinguals has shown that first language words are less likely to "blink" than second language words (e.g., Maier & Rahman, 2018). A third general motivation for applied studies has been to assess factors that affect control over attentional selection. For example, some studies reveal that mediation and engagement in other mindful tasks reduces the AB, an outcome interpreted to reflect better control over attention (e.g., Bailey et al., 2023; Y. Wang et al., 2023; Y. Wang, Xiao, et al., 2022). Social media has been particularly active on this point as discussed below.

The way in which social media and popular science have picked up on the AB has been unexpected, interesting, and humorous in some cases. An interesting example is Daniel Goleman, an American author and psychologist, explaining the AB to the Dalai Lama. In a YouTube clip, Goleman frames the explanation around the notion of the mind and its ability to focus on one thought to the exclusion of others, clearly referencing the Buddhist practice of meditation and how it relates to neuroscience. To the enjoyment of the Dalai Lama, Goleman relates the finding that, after 3 months of Vipassana meditation, there is a noticeable reduction in the AB (Slagter et al., 2007; van Vugt & Slagter, 2014). Goleman can also been seen with his coauthor Richard Davidson discussing their book "Altered Traits" on a YouTube video, which includes a few minutes of description of the AB and its everyday implications.

In another social media adoption of the AB, the magician Dom Twose performs a sleight of hand to make a coin appear and disappear, eventually attributing this to the "AB." It is not clear if he is serious or did this after being told about the phenomenon and chose to entertain his audience with a humorous reference to the psychological phenomenon. In either case, there were over 2200 views of this YouTube video as of 2017, making many nonscientists aware of naturally occurring, temporary failures in perceptual awareness.

Perhaps the greatest example of social media pickup of the AB is the "2-hour YouTube video by Andrew Huberman." Professor of Neurobiology and Ophthalmology at Stanford University, Huberman discusses attention and attention-based disorders. He covers various treatments, referring to the AB as a "tool" to ameliorate "potentially forever" attentional deficits because of a "lack of focus." Huberman acknowledges he is paraphrasing the book "Altered Traits" as referred to above, then explains the AB paradigm, attributing the attentional lapse to a deficit in neurochemistry. Huberman goes on to state that those with attention deficit

Table 2
Examples of Applied, Nonclinical Studies Using the Attentional Blink

Bilingualism (Colzato et al., 2008)	Driving (Hashemirad et al., 2023)
Forensics (sex offenders, lying; Beech et al. 2008; Bowman et al., 2014; Zappalà et al., 2016)	Linguistics (Cao et al., 2023; Maier & Rahman, 2018; Zhao et al., 2023)
Meditation (Bailey et al., 2023; Y. Wang, Xiao, et al., 2022; Y. Wang et al., 2023)	Music (Jefferies et al., 2008; Olivers & Nieuwenhuis, 2005)
Primates (Zylberberg et al., 2010)	Reading (de Groot et al., 2015; Visser, 2014)
Video advertising (Shen et al., 2020)	Video games (Green & Bavelier, 2003)

hyperactivity disorder have many more ABs than those without this condition (cf. Armstrong & Munoz, 2003) and corroborates the previously stated finding that those who practice "open monitoring" increase their "attentional focus." It is interesting to note that this video has over 6.8 M views (at the time of writing) meaning that the AB and attendant implications will have been disseminated to many people who do not necessarily read scientific publications. Other YouTube videos (e.g., "Master Your Focus Unleashing the Potential of Attentional Blinks") disseminate similar information.

This discussion of the social media uptake of the AB phenomenon provides an example of how scientific information about human cognition can become disseminated via social media. Clearly the accuracy, integrity, and interpretation of concepts such as the AB do not uphold the rigor of peer-reviewed journals, but the very presence of such topics on social media tells us about people's interest and appetite for understanding their own cognition. The fact that social media have picked up on this topic, which is arguably not the case with many scientific topics, underscores the impact of the AB outside the laboratory. But does this uptake serve a useful purpose? The answer to this question is both "yes" and "no." On the "no" side, if nonscientific readers attribute a phenomenon such as sleight of hand as exemplary of the AB, then clearly, they are mistaken and should be directed to read the literature on divided attention. On the other hand, if having a label for a psychological phenomenon causes the same readers to become curious and learn more about how this might apply to their own lives or work, then the argument can be made for a positive benefit both scientific, social, and economic. In other cases, such as Daniel Goleman offering an account of the AB to the Dalai Lama, it can be reasonably argued that there is utility in a highly influential spiritual leader and his followers becoming acquainted with a scientific phenomenon, even if the account given to the spiritual practice is not fully accurate.

Resistance to Suppression

One question remains to be addressed before we conclude. The large number of AB studies, only a small percentage of which have been discussed in the present report, could lead one to conclude a ubiquity of the phenomenon. The present authors think this is not far from the truth. Despite the few studies revealing either conditions or certain participant characteristics suggesting the AB can be suppressed, there is no good evidence in our opinion to come to this conclusion. Whereas Feinstein et al. (2004) concluded 50% of their participants showed no AB, the data to make this point are plotted as an average, thereby not allowing individual differences to be appreciated. Moreover, these "non blinkers," as they are referred, reveal T2 detection close to 100%, suggesting the possibility of a ceiling effect. The same criticism applies to reports by Kellie and Shapiro (2004), Shapiro, Caldwell, and Sorensen (1997) and Shapiro, Driver, et al. (1997) where in the former report participants who observed a "morphing" stimulus stream and in the latter looked for their own name as T2 were judged to not reveal an AB. Finally, although Martens et al. (2006) concluded 8% of the 207 participants tested in an AB paradigm were "nonblinkers," the authors write that each item in the stimulus stream was presented for 90 ms, implying no ISI and therefore an atypical AB stimulus stream. To conclude, although the above and perhaps other AB reports suggest conditions under which an AB may not occur, the veracity of this claim requires considerably more scrutiny than is possible within the confines of the present report.

Summary

The hundreds of published AB experiments, only a fraction of which have been discussed in the present report, speak to the ubiquity of the phenomenon under review here. Indeed, the knowledge gained from writing the present report has humbled the present authors. Importantly, the vantage point gained from this report has afforded us the perspective that the experimental findings, along with the formal and informal theories arising from them, mirror the methodological innovations and increased understanding of how the human brain works over the 32 years of AB research. We believe the readers of this report will concur with the authors that currently no single account of the AB can fully explain this phenomenon, as the AB taps multiple cognitive demands, any one of which will affect its outcome. Challenges for future research concern the development of a unifying theory; addressing questions that link AB effects with learning, motivation, and context; advancing understanding of how brain mechanisms operate in orchestra to edit awareness; and, indeed, gaining further understanding of the nature, depth, and temporality of human awareness. These large questions bring us full circle to account for why the AB has inspired so many talented researchers over so many years.

We found the writing of this report to be an enjoyable exercise as it has given us an opportunity to reflect on where the original 1992 report has taken us and the field. We hasten to reiterate that this report should not be viewed as an encompassing review of the AB, but more of a retrospective "helicopter" view of what has transpired in 32 years of interest in the phenomenon. As instructed by *Journal of Experimental Psychology: Human Perception and Performance*, we intend the report to reflect the impact, reach, and major developments, and hope we have accomplished that goal. We are convinced the AB has attracted the interest of so many researchers because of its encompassing multiple cognitive subsystems, including encoding, attention, short- and long- term memory, and report.

We are indebted to the *Journal of Experimental Psychology: Human Perception and Performance* for the opportunity to write this report and to help the journal celebrate a half century of being a pillar of experimental psychology broadly, and cognitive psychology and cognitive neuroscience more specifically. We are struck by how many and varied are the scientists who have been influenced by the AB and who have contributed hugely to its dissemination. As mentioned in the introduction, we are also struck by how the study of this phenomenon has mirrored and perhaps even encouraged the approaches, methods, and technical developments since 1992. We look forward to future technological advances that will enable brain states to be registered with temporal precision, such as optically pumped magnetometers, and the emerging ways to precisely stimulate the brain including transcranial electrical and magnetic stimulation, ultrasound, photobiomodulation, and optogenetics and magnetogenetics, to see where they will take the study of the AB and the field at large.

References

Amador-Campos, J. A., Aznar-Casanova, J. A., Bezerra, I., Torro-Alves, N., & Sánchez, M. M. (2015). Attentional blink in children with attention deficit hyperactivity disorder. *Brazilian Journal of Psychiatry*, *37*(2), 133–138. https://doi.org/10.1590/1516-4446-2014-1415

Anderson, A. K., & Phelps, E. A. (2001). Lesions of the human amygdala impair enhanced perception of emotionally salient events. *Nature*, *411*(6835), 305–309. https://doi.org/10.1038/35077083

Armstrong, I. T., & Munoz, D. P. (2003). Attentional blink in adults with attention-deficit hyperactivity disorder. Influence of eye movements. *Experimental Brain Research Experimentelle Hirnforschung Expérimentation Cérébrale*, *152*(2), 243–250. https://doi.org/10.1007/s00221-003-1535-0

Arnell, K. M. (2006). Visual, auditory, and cross-modality dual-task costs: Electrophysiological evidence for an amodal bottleneck on working memory consolidation. *Perception & Psychophysics*, *68*(3), 447–457. https://doi.org/10.3758/bf03193689

Arnell, K. M., & Jolicœur, P. (1999). The attentional blink across stimulus modalities: Evidence for central processing limitations. *Journal of Experimental Psychology: Human Perception and Performance*, *25*(3), 630–648. https://doi.org/10.1037/0096-1523.25.3.630

Arnell, K. M., & Larson, J. M. (2002). Cross-modality attentional blinks without preparatory task-set switching. *Psychonomic Bulletin & Review*, *9*(3), 497–506. https://doi.org/10.3758/bf03196305

Arnell, K. M., & Shapiro, K. L. (2011). Attentional blink and repetition blindness. *Wiley Interdisciplinary Reviews: Cognitive Science*, *2*(3), 336–344. https://doi.org/10.1002/wcs.129

Arumäe, K., Kreegipuu, K., & Vainik, U. (2019). Assessing the overlap between three measures of food reward. *Frontiers in Psychology*, *10*, Article 883. https://doi.org/10.3389/fpsyg.2019.00883

Badcock, N. A., Hogben, J. H., & Fletcher, J. F. (2008). No differential attentional blink in dyslexia after controlling for baseline sensitivity. *Vision Research*, *48*(13), 1497–1502. https://doi.org/10.1016/j.visres.2008.03.008

Bailey, N. W., Baell, O., Payne, J. E., Humble, G., Geddes, H., Cahill, I., Hill, A. T., Chung, S. W., Emonson, M., Murphy, O. W., & Fitzgerald, P. B. (2023). Experienced meditators show multifaceted attention-related differences in neural activity. *Mindfulness*, *14*(11), 2670–2698. https://doi.org/10.1007/s12671-023-02224-2

Beanland, V., & Pammer, K. (2012). Minds on the blink: The relationship between inattentional blindness and attentional blink. *Attention, Perception, & Psychophysics*, *74*(2), 322–330. https://doi.org/10.3758/s13414-011-0241-4

Beech, A. R., Kalmus, E., Tipper, S. P., Baudouin, J. Y., Flak, V., & Humphreys, G. W. (2008). Children induce an enhanced attentional blink in child molesters. *Psychological Assessment*, *20*(4), 397–402. https://doi.org/10.1037/a0013587

Bowman, H., Filetti, M., Alsufyani, A., Janssen, D., & Su, L. (2014). Countering countermeasures: Detecting identity lies by detecting conscious breakthrough. *PLoS ONE*, *9*(3), Article e90595. https://doi.org/10.1371/journal.pone.0090595

Bowman, H., Jones, W., Pincham, H., Fleming, S., Cleeremans, A., & Smith, M. (2022). Modelling the simultaneous encoding/serial experience theory of the perceptual moment: A blink of meta-experience. *Neuroscience of Consciousness*, *2022*(1), Article niac003. https://doi.org/10.1093/nc/niac003

Bowman, H., & Wyble, B. (2007). The simultaneous type, serial token model of temporal attention and working memory. *Psychological Review*, *114*(1), 38–70. https://doi.org/10.1037/0033-295x.114.1.38

Brehaut, J. C., Enns, J. T., & Di Lollo, V. (1999). Visual masking plays two roles in the attentional blink. *Perception & Psychophysics*, *61*(7), 1436–1448. https://doi.org/10.3758/bf03206192

Broadbent, D. E., & Broadbent, M. H. P. (1987). From detection to identification: Response to multiple targets in rapid serial visual presentation. *Perception & Psychophysics*, *42*(2), 105–113. https://doi.org/10.3758/bf03210498

Buchholz, J., & Davies, A. A. (2007). Attentional blink deficits observed in dyslexia depend on task demands. *Vision Research*, *47*(10), 1292–1302. https://doi.org/10.1016/j.visres.2006.11.028

Campbell, D. W., Stewart, S., Gray, C. E. P., Ryan, C. L., Fettes, P., McLandress, A. J., & Dell'Aquila, R. (2018). Chronic cannabis use and attentional bias: Extended attentional capture to cannabis cues. *Addictive Behaviors*, *81*(3), 17–21. https://doi.org/10.1016/j.addbeh.2018.01.008

Cao, H.-W., Zhang, E.-H., Xiang, X., Li, D., & Lei, V. L. C. (2023). Lexico-semantic activation of translation equivalents during the attentional blink. *Journal of Psycholinguistic Research*, *52*(1), 101–113. https://doi.org/10.1007/s10936-021-09820-0

Chun, M. M. (1997). Types and tokens in visual processing: A double dissociation between the attentional blink and repetition blindness. *Journal of Experimental Psychology: Human Perception and Performance*, *23*(3), 738–755. https://doi.org/10.1037/0096-1523.23.3.738

Chun, M. M., & Potter, M. C. (1995). A two-stage model for multiple target detection in rapid serial visual presentation. *Journal of Experimental Psychology: Human Perception and Performance*, *21*(1), 109–127. https://doi.org/10.1037/0096-1523.21.1.109

Colzato, L. S., Bajo, M. T., van den Wildenberg, W., Paolieri, D., Nieuwenhuis, S., Heij, W. L., & Hommel, B. (2008). How does bilingualism improve executive control? A comparison of active and reactive inhibition mechanisms. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, *34*(2), 302–312. https://doi.org/10.1037/0278-7393.34.2.302

Correani, A., & Humphreys, G. W. (2011). An impaired attentional dwell time after parietal and frontal lesions related to impaired selective attention not unilateral neglect. *Cognitive Neuropsychology*, *28*(5), 363–385. https://doi.org/10.1080/02643294.2012.656587

de Groot, B. J. A., van den Bos, K. P., van der Meulen, B. F., & Minnaert, A. E. M. G. (2015). The attentional blink in typically developing and reading-disabled children. *Journal of Experimental Child Psychology*, *139*, 51–70. https://doi.org/10.1016/j.jecp.2015.05.003

Dehaene, S., Changeux, J.-P., Naccache, L., Sackur, J., & Sergent, C. (2006). Conscious, preconscious, and subliminal processing: A testable taxonomy. *Trends in Cognitive Sciences*, *10*(5), 204–211. https://doi.org/10.1016/j.tics.2006.03.007

Dell'Acqua, R., Turatto, M., & Jolicoeur, P. (2001). Cross-modal attentional deficits in processing tactile stimulation. *Perception & Psychophysics*, *63*(5), 777–789. https://doi.org/10.3758/bf03194437

Denison, R. N., Carrasco, M., & Heeger, D. J. (2021). A dynamic normalization model of temporal attention. *Nature Human Behaviour*, *5*(12), 1674–1685. https://doi.org/10.1038/s41562-021-01129-1

DePalma, F. M., Ceballos, N., & Graham, R. (2017). Attentional blink to alcohol cues in binge drinkers versus non-binge drinkers. *Addictive Behaviors*, *73*, 67–73. https://doi.org/10.1016/j.addbeh.2017.04.020

Di Lollo, V. D., Kawahara, J., Ghorashi, S. M. S., & Enns, J. T. (2005). The attentional blink: Resource depletion or temporary loss of control? *Psychological Research*, *69*(3), 191–200. https://doi.org/10.1007/s00426-004-0173-x

Donnadieu, S., Berger, C., Lallier, M., Marendaz, C., & Laurent, A. (2015). Is the impairment in temporal allocation of visual attention in children with ADHD related to a developmental delay or a structural cognitive deficit? *Research in Developmental Disabilities*, *36*, 384–395. https://doi.org/10.1016/j.ridd.2014.10.014

Drew, T., & Shapiro, K. (2006). Representational masking and the attentional blink. *Visual Cognition*, *13*(4), 513–528. https://doi.org/10.1080/13506280544000255

Duncan, J., & Humphreys, G. W. (1989). Visual search and stimulus similarity. *Psychological Review*, *96*(3), 433–458. https://doi.org/10.1037/0033-295X.96.3.433

Duncan, J., Martens, S., & Ward, R. (1997). Restricted attentional capacity within but not between sensory modalities. *Nature*, *387*(6635), 808–810. https://doi.org/10.1038/42947

Duncan, J., Ward, R., & Shapiro, K. (1994). Direct measurement of attentional dwell time in human vision. *Nature*, *369*(6478), 313–315. https://doi.org/10.1038/369313a0

Dux, P. E., & Marois, R. (2007). Repetition blindness is immune to the central bottleneck. *Psychonomic Bulletin & Review*, *14*(4), 729–734. https://doi.org/10.3758/bf03196829

Dux, P. E., & Marois, R. (2009). The attentional blink: A review of data and theory. *Attention, Perception, & Psychophysics*, *71*(8), 1683–1700. https://doi.org/10.3758/app.71.8.1683

English, M. C. W., Maybery, M. T., & Visser, T. A. W. (2017). Threatening faces fail to guide attention for adults with autistic-like traits. *Autism Research*, *10*(2), 311–320. https://doi.org/10.1002/aur.1658

Facoetti, A., Ruffino, M., Peru, A., Paganoni, P., & Chelazzi, L. (2008). Sluggish engagement and disengagement of non-spatial attention in dyslexic children. *Cortex*, *44*(9), 1221–1233. https://doi.org/10.1016/j.cortex.2007.10.007

Farr, Z., Broomfield, N. M., & Coventry, K. R. (2024). A systematic review of attentional bias in problem gambling. *Journal of Gambling Studies*, *40*(2), 493–519. https://doi.org/10.1007/s10899-023-10260-9

Feinstein, J. S., Stein, M. B., Castillo, G. N., & Paulus, M. P. (2004). From sensory processes to conscious perception. *Consciousness and Cognition*, *13*(2), 323–335. https://doi.org/10.1016/j.concog.2003.10.004

Folk, C. L., Leber, A. B., & Egeth, H. E. (2002). Made you blink! Contingent attentional capture produces a spatial blink. *Perception & Psychophysics*, *64*(5), 741–753. https://doi.org/10.3758/bf03194741

Gaigg, S. B., & Bowler, D. M. (2009). Brief report: Attenuated emotional suppression of the attentional blink in autism spectrum disorder: Another non-social abnormality? *Journal of Autism and Developmental Disorders*, *39*(8), 1211–1217. https://doi.org/10.1007/s10803-009-0719-2

Giesbrecht, B., Bischof, W. F., & Kingstone, A. (2003). Visual masking during the attentional blink: Tests of the object substitution hypothesis. *Journal of Experimental Psychology: Human Perception and Performance*, *29*(1), 238–258. https://doi.org/10.1037/0096-1523.29.1.238

Giesbrecht, B., & Di Lollo, V. (1998). Beyond the attentional blink: Visual masking by object substitution. *Journal of Experimental Psychology: Human Perception and Performance*, *24*(5), 1454–1466. https://doi.org/10.1037/0096-1523.24.5.1454

Green, C. S., & Bavelier, D. (2003). Action video game modifies visual selective attention. *Nature*, *423*(6939), 534–537. https://doi.org/10.1038/nature01647

Gross, J., Schmitz, F., Schnitzler, I., Kessler, K., Shapiro, K., Hommel, B., & Schnitzler, A. (2004). Modulation of long-range neural synchrony reflects temporal limitations of visual attention in humans. *Proceedings of the National Academy of Sciences*, *101*(35), 13050–13055. https://doi.org/10.1073/pnas.0404944101

Hashemirad, S. R., Vaziri-Pashkam, M., & Abbaszadeh, M. (2023). *Time course of dual-task interference in the brain in simulated driving environment*. BioRxiv. https://doi.org/10.1101/2023.11.06.565914

Hillstrom, A. P., Shapiro, K. L., & Spence, C. (2002). Attentional limitations in processing sequentially presented vibrotactile targets. *Perception & Psychophysics*, *64*(7), 1068–1082. https://doi.org/10.3758/bf03194757

Huang, Y. M., Baddeley, A., & Young, A. W. (2008). Attentional capture by emotional stimuli is modulated by semantic processing. *Journal of Experimental Psychology: Human Perception and Performance*, *34*(2), 328–339. https://doi.org/10.1037/0096-1523.34.2.328

Jackson, M. C., & Raymond, J. E. (2006). The role of attention and familiarity in face identification. *Perception & Psychophysics*, *68*(4), 543–557. https://doi.org/10.3758/BF03208757

Jefferies, L. N., Smilek, D., Eich, E., & Enns, J. T. (2008). Emotional valence and arousal interact in attentional control. *Psychological Science*, *19*(3), 290–295. https://doi.org/10.1111/j.1467-9280.2008.02082.x

Jiang, Y., Tian, Y., & Wang, K. (2013). The change of attentional blink and repetition blindness after cerebellar lesions. *Journal of Clinical Neuroscience*, *20*(12), 1742–1746. https://doi.org/10.1016/j.jocn.2013.01.022

Jimenez, A. M., Lee, J., Wynn, J. K., Cohen, M. S., Engel, S. A., Glahn, D. C., Nuechterlein, K. H., Reavis, E. A., & Green, M. F. (2016). *Abnormal ventral and dorsal attention network activity during single and dual target detection in schizophrenia*. https://doi.org/10.3389/fpsyg.2016.00323

Kan, Y., Duan, H., Chen, X., Wang, X., Xue, W., & Hu, W. (2019). Attentional blink affected by acute stress in women: The role of affective stimuli and attentional resources. *Consciousness and Cognition*, *75*, Article 102796. https://doi.org/10.1016/j.concog.2019.102796

Kawahara, J.-I., Enns, J. T., & Di Lollo, V. D. (2006). The attentional blink is not a unitary phenomenon. *Psychological Research*, *70*(6), 405–413. https://doi.org/10.1007/s00426-005-0007-5

Kawahara, J.-I., Kumada, T., & Di Lollo, V. (2006). The attentional blink is governed by a temporary loss of control. *Psychonomic Bulletin & Review*, *13*(5), 886–890. https://doi.org/10.3758/bf03194014

Kellie, F. J., & Shapiro, K. L. (2004). Object file continuity predicts attentional blink magnitude. *Perception & Psychophysics*, *66*(4), 692–712. https://doi.org/10.3758/bf03194912

Kimata, A. R., Zheng, B., Watanabe, T., & Asaad, W. F. (2023). The temporal cost of deploying attention limits accurate target identification in rapid serial visual presentation. *Scientific Reports*, *13*(1), Article 3590. https://doi.org/10.1038/s41598-023-30748-z

Klein, C., Arend, I. C., Beauducel, A., & Shapiro, K. L. (2011). Individuals differ in the attentional blink: Mental speed and intra-subject stability matter. *Intelligence*, *39*(1), 27–35. https://doi.org/10.1016/j.intell.2010.11.004

Koelewijn, T., Van der Burg, E., Bronkhorst, A., & Theeuwes, J. (2008). Priming T2 in a visual and auditory attentional blink task. *Perception & Psychophysics*, *70*(4), 658–666. https://doi.org/10.3758/pp.70.4.658

Kouider, S., & Dehaene, S. (2007). Levels of processing during non-conscious perception: A critical review of visual masking. *Philosophical Transactions of the Royal Society B: Biological Sciences*, *362*(1481), 857–875. https://doi.org/10.1098/rstb.2007.2093

Krans, J., Reinecke, A., de Jong, P. J., Näring, G., & Becker, E. S. (2012). Analogue trauma results in enhanced encoding of threat information at the expense of neutral information. *Journal of Anxiety Disorders*, *26*(6), 656–664. https://doi.org/10.1016/j.janxdis.2012.05.003

Kreddig, N., Hasenbring, M. I., & Keogh, E. (2022). Comparing the effects of thought suppression and focused distraction on pain-related attentional biases in men and women. *The Journal of Pain*, *23*(11), 1958–1972. https://doi.org/10.1016/j.jpain.2022.07.004

Kristjánsson, Á., & Nakayama, K. (2002). The attentional blink in space and time. *Vision Research*, *42*(17), 2039–2050. https://doi.org/10.1016/s0042-6989(02)00129-3

Li, K., Russell, C., Balaji, N., Saleh, Y., Soto, D., & Malhotra, P. A. (2016). The effects of motivational reward on the pathological attentional blink following right hemisphere stroke. *Neuropsychologia*, *92*, 190–196. https://doi.org/10.1016/j.neuropsychologia.2016.03.037

Lindh, D., Sligte, I. G., Assecondi, S., Shapiro, K. L., & Charest, I. (2019). Conscious perception of natural images is constrained by category-related visual features. *Nature Communications*, *10*(1), Article 4106. https://doi.org/10.1038/s41467-019-12135-3

Linkenkaer-Hansen, K., Nikouline, V. V., Palva, J. M., & Ilmoniemi, R. J. (2001). Long-range temporal correlations and scaling behavior in human brain oscillations. *The Journal of Neuroscience*, *21*(4), 1370–1377. https://doi.org/10.1523/jneurosci.21-04-01370.2001

Lu, M., Lian, C., & Zeng, X. (2024). An attentional blink research on different types of words in male with substance use disorder. *International Journal of Mental Health and Addiction*, *22*(1), 494–512. https://doi.org/10.1007/s11469-022-00885-z

Luck, S. J., Vogel, E. K., & Shapiro, K. L. (1996). Word meanings can be accessed but not reported during the attentional blink. *Nature*, *383*(6601), 616–618. https://doi.org/10.1038/383616a0

Maier, M., & Rahman, R. A. (2018). Native language promotes access to visual consciousness. *Psychological Science*, *29*(11), 1757–1772. https://doi.org/10.1177/0956797618782181

Marois, R., Yi, D.-J., & Chun, M. M. (2004). The neural fate of consciously perceived and missed events in the attentional blink. *Neuron*, *41*(3), 465–472. https://doi.org/10.1016/s0896-6273(04)00012-1

Martens, S., Munneke, J., Smid, H., & Johnson, A. (2006). Quick minds don't blink: Electrophysiological correlates of individual differences in attentional selection. *Journal of Cognitive Neuroscience*, *18*(9), 1423–1438. https://doi.org/10.1162/jocn.2006.18.9.1423

Marti, S., Sigman, M., & Dehaene, S. (2012). A shared cortical bottleneck underlying attentional blink and psychological refractory period. *NeuroImage*, *59*(3), 2883–2898. https://doi.org/10.1016/j.neuroimage.2011.09.063

Mathewson, K. J., Arnell, K. M., & Mansfield, C. A. (2008). Capturing and holding attention: The impact of emotional words in rapid serial visual presentation. *Memory & Cognition*, *36*(1), 182–200. https://doi.org/10.3758/MC.36.1.182

McHugo, M., Olatunji, B. O., & Zald, D. H. (2013). The emotional attentional blink: What we know so far. *Frontiers in Human Neuroscience*, *7*, Article 151. https://doi.org/10.3389/fnhum.2013.00151

McLaughlin, E. N., Shore, D. I., & Klein, R. M. (2001). The attentional blink is immune to masking-induced data limits. *The Quarterly Journal of Experimental Psychology Section A*, *54*(1), 169–196. https://doi.org/10.1080/02724980042000075

Mondor, T. A. (1998). A transient processing deficit following selection of an auditory target. *Psychonomic Bulletin & Review*, *5*(2), 305–311. https://doi.org/10.3758/bf03212956

Neimeijer, R. A. M., Roefs, A., & de Jong, P. J. (2017). Heightened attentional capture by visual food stimuli in anorexia nervosa. *Journal of Abnormal Psychology*, *126*(6), 805–811. https://doi.org/10.1037/abn0000275

Nieuwenhuis, S., Gilzenrat, M. S., Holmes, B. D., & Cohen, J. D. (2005). The role of the locus coeruleus in mediating the attentional blink: A neurocomputational theory. *Journal of Experimental Psychology: General*, *134*(3), 291–307. https://doi.org/10.1037/0096-3445.134.3.291

Nijhof, A. D., Shapiro, K. L., Catmur, C., & Bird, G. (2020). No evidence for a common self-bias across cognitive domains. *Cognition*, *197*, Article 104186. https://doi.org/10.1016/j.cognition.2020.104186

Nijhof, A. D., von Trott zu Solz, J., Catmur, C., & Bird, G. (2022). Equivalent own name bias in autism: An EEG study of the Attentional Blink. *Cognitive, Affective, & Behavioral Neuroscience*, *22*(3), 625–639. https://doi.org/10.3758/s13415-021-00967-w

Olatunji, B. O. (2021). Emotional induced attentional blink in obsessive-compulsive disorder. *Journal of Affective Disorders*, *283*(3), 101–107. https://doi.org/10.1016/j.jad.2020.12.204

Olatunji, B. O., Liu, Q., Zald, D. H., & Cole, D. A. (2022). Emotional induced attentional blink in trauma-exposed veterans: Associations with trauma specific and nonspecific symptoms. *Journal of Anxiety Disorders*, *87*, Article 102541. https://doi.org/10.1016/j.janxdis.2022.102541

Olivers, C. N. L., & Meeter, M. (2008). A boost and bounce theory of temporal attention. *Psychological Review*, *115*(4), 836–863. https://doi.org/10.1037/a0013395

Olivers, C. N. L., & Nieuwenhuis, S. (2005). The beneficial effect of concurrent task-irrelevant mental activity on temporal attention. *Psychological Science*, *16*(4), 265–269. https://doi.org/10.1111/j.0956-7976.2005.01526.x

Olivers, C. N. L., van der Stigchel, S., & Hulleman, J. (2007). Spreading the sparing: Against a limited-capacity account of the attentional blink. *Psychological Research*, *71*(2), 126–139. https://doi.org/10.1007/s00426-005-0029-z

Perry, R. J., & Hodges, J. R. (2003). Dissociation between top-down attentional control and the time course of visual attention as measured by attentional dwell time in patients with mild cognitive impairment. *European Journal of Neuroscience*, *18*(2), 221–226. https://doi.org/10.1046/j.1460-9568.2003.02754.x

Posner, M. I. (1980). Orienting of Attention*. *Quarterly Journal of Experimental Psychology*, *32*(1), 3–25. https://doi.org/10.1080/00335558008248231

Potter, M. C., Chun, M. M., Banks, B. S., & Muckenhoupt, M. (1998). Two attentional deficits in serial target search: The visual attentional blink and an amodal task-switch deficit. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, *24*(4), 979–992. https://doi.org/10.1037/0278-7393.24.4.979

Raymond, J. E. (2003). New objects, not new features, trigger the attentional blink. *Psychological Science*, *14*(1), 54–59. https://doi.org/10.1111/1467-9280.01418

Raymond, J. E., & O'Brien, J. L. (2009). Selective visual attention and motivation: The consequences of value learning in an attentional blink task. *Psychological Science*, *20*(8), 981–988. https://doi.org/10.1111/j.1467-9280.2009.02391.x

Raymond, J. E., Shapiro, K. L., & Arnell, K. M. (1992). Temporary suppression of visual processing in an RSVP task: An attentional blink? *Journal of Experimental Psychology: Human Perception and Performance*, *18*(3), 849–860. https://doi.org/10.1037/0096-1523.18.3.849

Raymond, J. E., Shapiro, K. L., & Arnell, K. M. (1995). Similarity determines the Attentional Blink. *Journal of Experimental Psychology: Human Performance and Perception*, *21*(3), 653–662. https://doi.org/10.1037/0096-1523.21.3.653

Schmitz, F., Naumann, E., Biehl, S., & Svaldi, J. (2015). Gating of attention towards food stimuli in binge eating disorder. *Appetite*, *95*, 368–374. https://doi.org/10.1016/j.appet.2015.07.023

Schwabe, L., & Wolf, O. T. (2010). Emotional modulation of the attentional blink: Is there an effect of stress? *Emotion*, *10*(2), 283–288. https://doi.org/10.1037/a0017751

Seiffert, A., & Di Lollo, V. (1997). Low-level masking in the attentional blink. *Journal of Experimental Psychology: Human Perception & Performance*, *23*(4), 1061–1073. https://doi.org/10.1037/0096-1523.23.4.1061

Shapiro, K. L., Caldwell, J. I., & Sorensen, R. E. (1997). Personal names and the attentional blink: The 'cocktail party' effect revisited. *Journal of Experimental Psychology: Human Perception and Performance*, *23*(2), 504–514. https://doi.org/10.1037/0096-1523.23.2.504

Shapiro, K. L., Driver, J., Ward, R., & Sorensen, R. E. (1997). Priming from the attentional blink: A failure to extract visual tokens but not visual types. *Psychological Science*, *8*(2), 95–100. https://doi.org/10.1111/j.1467-9280.1997.tb00689.x

Shapiro, K. L., Hanslmayr, S., Enns, J. T., & Lleras, A. (2017). Alpha, beta: The rhythm of the attentional blink. *Psychonomic Bulletin & Review*, *24*(6), 1862–1869. https://doi.org/10.3758/s13423-017-1257-0

Shapiro, K. L., Johnston, S. J., Vogels, W., Zaman, A., & Roberts, N. (2007). Increased functional magnetic resonance imaging activity during nonconscious perception in the attentional blink. *NeuroReport*, *18*(4), 341–345. https://doi.org/10.1097/wnr.0b013e32801299e2

Shapiro, K. L., Raymond, J. E., & Arnell, K. M. (1994). Attention to visual pattern information produces the attentional blink in rapid serial visual presentation. *Journal of Experimental Psychology: Human Perception and Performance*, *20*(2), 357–371. https://doi.org/10.1037/0096-1523.20.2.357

Shen, W., Gu, H., Ball, L. J., Yuan, Y., Yu, C., Shi, R., & Huang, T. (2020). The impact of advertising creativity, warning-based appeals and green dispositions on the attentional effectiveness of environmental advertisements. *Journal of Cleaner Production*, *271*, Article 122618. https://doi.org/10.1016/j.jclepro.2020.122618

Sheppard, D. M., Duncan, J., Shapiro, K. L., & Hillstrom, A. P. (2002). Objects and events in the attentional blink. *Psychological Science*, *13*(5), 410–415. https://doi.org/10.1111/1467-9280.00473

Singh, D., & Sunny, M. M. (2023). Spatial distribution of emotional attentional blink under top-down attentional control. *Cognitive Processing*, *24*(1), 153–159. https://doi.org/10.1007/s10339-022-01109-x

Slagter, H. A., Lutz, A., Greischar, L. L., Francis, A. D., Nieuwenhuis, S., Davis, J. M., & Davidson, R. J. (2007). Mental training affects distribution of limited brain resources. *PLoS Biology*, *5*(6), Article e138. https://doi.org/10.1371/journal.pbio.0050138

Slagter, H. A., van Wouwe, N. C., Kanoff, K., Grasman, R. P. P. P., Claassen, D. O., van den Wildenberg, W. P. M., & Wylie, S. A. (2016). Dopamine and temporal attention: An attentional blink study in

Parkinson's disease patients on and off medication. *Neuropsychologia*, *91*, 407–414. https://doi.org/10.1016/j.neuropsychologia.2016.09.006

Soto-Faraco, S., Spence, C., Fairbank, K., Kingstone, A., Hillstrom, A. P., & Shapiro, K. (2002). A crossmodal attentional blink between vision and touch. *Psychonomic Bulletin & Review*, *9*(4), 731–738. https://doi.org/10.3758/BF03196328

Strauss, G. P., Catalano, L. T., Llerena, K., & Gold, J. M. (2013). The processing of emotional stimuli during periods of limited attentional resources in schizophrenia. *Journal of Abnormal Psychology*, *122*(2), 492–505. https://doi.org/10.1037/a0031212

Su, L., Wyble, B., Zhou, L., Wang, K., Wang, Y., Cheung, E. F. C., Bowman, H., & Chan, R. C. K. (2015). Temporal perception deficits in schizophrenia: Integration is the problem, not deployment of attentions. *Scientific Reports*, *5*(1), Article 9745. https://doi.org/10.1038/srep09745

Tononi, G., & Koch, C. (2008). The neural correlates of consciousness. *Annals of the New York Academy of Sciences*, *1124*(1), 239–261. https://doi.org/10.1196/annals.1440.004

Treisman, A. M., & Gelade, G. (1980). A feature-integration theory of attention. *Cognitive Psychology*, *12*(1), 97–136. https://doi.org/10.1016/0010-0285(80)90005-5

Tremblay, S., Vachon, F., & Jones, D. M. (2005). Attentional and perceptual sources of the auditory attentional blink. *Perception & Psychophysics*, *67*(2), 195–208. https://doi.org/10.3758/bf03206484

Trippe, R. H., Hewig, J., Heydel, C., Hecht, H., & Miltner, W. H. R. (2007). Attentional Blink to emotional and threatening pictures in spider phobics: Electrophysiology and behavior. *Brain Research*, *1148*, 149–160. https://doi.org/10.1016/j.brainres.2007.02.035

Vachon, F., & Tremblay, S. (2005). Auditory attentional blink: Masking the second target is necessary, delayed masking is sufficient. *Canadian Journal of Experimental Psychology*, *59*(4), 279–286. https://doi.org/10.1037/h0087480

van Dijk, H., Schoffelen, J.-M., Oostenveld, R., & Jensen, O. (2008). Prestimulus oscillatory activity in the alpha band predicts visual discrimination ability. *The Journal of Neuroscience*, *28*(8), 1816–1823. https://doi.org/10.1523/jneurosci.1853-07.2008

van Vugt, M. K., & Slagter, H. A. (2014). Control over experience? Magnitude of the attentional blink depends on meditative state. *Consciousness and Cognition*, *23*, 32–39. https://doi.org/10.1016/j.concog.2013.11.001

Vila-Castelar, C., Ly, J. J., Kaplan, L., Van Dyk, K., Berger, J. T., Macina, L. O., Stewart, J. L., & Foldi, N. S. (2019). Attention measures of accuracy, variability, and fatigue detect early response to donepezil in Alzheimer's disease: A randomized, double-blind, placebo-controlled pilot trial. *Archives of Clinical Neuropsychology*, *34*(3), 277–289. https://doi.org/10.1093/arclin/acy032

Visser, T. A. W. (2014). Evidence for deficits in the temporal attention span of poor readers. *PLoS ONE*, *9*(3), Article e91278. https://doi.org/10.1371/journal.pone.0091278

Visser, T. A. W. (2015). Expectancy-based modulations of Lag-1 sparing and extended sparing during the attentional blink. *Journal of Experimental Psychology: Human Perception and Performance*, *41*(2), 462–478. https://doi.org/10.1037/a0038754

Walsh-Messinger, J., Ramirez, P. M., Wong, P., Antonius, D., Aujero, N., McMahon, K., Opler, L. A., & Malaspina, D. (2014). Impairment in emotional modulation of attention and memory in schizophrenia. *Schizophrenia Research*, *157*(1–3), 63–69. https://doi.org/10.1016/j.schres.2014.05.014

Wang, J., Luo, Y., Aleman, A., & Martens, S. (2022). Training the attentional blink: Subclinical depression decreases learning potential. *Psychological Research*, *86*(6), 1980–1995. https://doi.org/10.1007/s00426-021-01603-5

Wang, Y., Chen, Y., Sun, Y., Zhang, K., Wang, N., Sun, Y., Lin, X., Wang, J., & Luo, F. (2023). Gender differences in the benefits of meditation training on attentional blink. *Current Psychology*, *42*(7), 5178–5187. https://doi.org/10.1007/s12144-021-01891-5

Wang, Y., Xiao, L., Gong, W., Chen, Y., Lin, X., Sun, Y., Wang, N., Wang, J., & Luo, F. (2022). Mindful non-reactivity is associated with improved accuracy in attentional blink testing: A randomized controlled trial. *Current Psychology*, *41*(12), 8908–8920. https://doi.org/10.1007/s12144-021-01377-4

Ward, R., Duncan, J., & Shapiro, K. (1996). The slow time-course of visual attention. *Cognitive Psychology*, *30*(1), 79–109. https://doi.org/10.1006/cogp.1996.0003

Waters, A. J., Heishman, S. J., Lerman, C., & Pickworth, W. (2007). Enhanced identification of smoking-related words during the attentional blink in smokers. *Addictive Behaviors*, *32*(12), 3077–3082. https://doi.org/10.1016/j.addbeh.2007.05.016

Weichselgartner, E., & Sperling, G. (1987). Dynamics of automatic and controlled visual attention. *Science*, *238*(4828), 778–780. https://doi.org/10.1126/science.3672124

Woltering, S., Chen, S., & Jia, Y. (2021). Neural correlates of attentional bias to food stimuli in obese adolescents. *Brain Topography*, *34*(2), 182–191. https://doi.org/10.1007/s10548-020-00812-2

Wyble, B., Bowman, H., & Nieuwenstein, M. (2009). The attentional blink provides episodic distinctiveness: Sparing at a cost. *Journal of Experimental Psychology: Human Perception and Performance*, *35*(3), 787–807. https://doi.org/10.1037/a0013902

Wyble, B., Potter, M. C., Bowman, H., & Nieuwenstein, M. (2011). Attentional episodes in visual perception. *Journal of Experimental Psychology: General*, *140*(3), 488–505. https://doi.org/10.1037/a0023612

Xu, L., Ma, X., Zhao, W., Luo, L., Yao, S., & Kendrick, K. M. (2015). Oxytocin enhances attentional bias for neutral and positive expression faces in individuals with higher autistic traits. *Psychoneuroendocrinology*, *62*, 352–358. https://doi.org/10.1016/j.psyneuen.2015.09.002

Zappalà, A., Antfolk, J., Dombert, B., Mokros, A., & Santtila, P. (2016). Identifying deviant sexual interest in a sex offender sample using dual-target rapid serial visual presentation task. *The Journal of Forensic Psychiatry & Psychology*, *27*(2), 281–307. https://doi.org/10.1080/14789949.2015.1122820

Zhang, D., Zhang, R., Zhou, L., Zhou, K., & Chang, C. (2023). The brain network underlying attentional blink predicts symptoms of attention deficit hyperactivity disorder in children. *Cerebral Cortex*, *33*(6), 2761–2773. https://doi.org/10.1093/cercor/bhac240

Zhao, S., Zhou, Y., Ma, F., Xie, J., Feng, C., & Feng, W. (2023). The dissociation of semantically congruent and incongruent cross-modal effects on the visual attentional blink. *Frontiers in Neuroscience*, *17*, Article 1295010. https://doi.org/10.3389/fnins.2023.1295010

Zivony, A., & Lamy, D. (2016). Attentional capture and engagement during the attentional blink: A "camera" metaphor of attention. *Journal of Experimental Psychology: Human Perception and Performance*, *42*(11), 1886–1902. https://doi.org/10.1037/xhp0000286

Zylberberg, A., Slezak, D. F., Roelfsema, P. R., Dehaene, S., & Sigman, M. (2010). The brain's router: A cortical network model of serial processing in the primate brain. *PLoS Computational Biology*, *6*(4), Article e1000765. https://doi.org/10.1371/journal.pcbi.1000765

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